A Predictive Framework for Fundamental Constants, Neutrino Masses, and Cosmic Tensions Based on a Discrete Spacetime Lattice with Octahedral Symmetry

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We present a predictive phenomenological framework—Wen Dafang Theory (WDF)—in which fundamental physical constants and cosmic phenomena emerge from a minimal set of axioms: the underlying spacetime is a discrete lattice with $O_h = 48$ octahedral symmetry, the electron mass $m_e \equiv 1$ serves as the mass benchmark, and all physical interactions correspond to k-form dimensional protocols for organizing information on the lattice. From the single action S_{WDF}, we derive: (i) the proton-to-electron mass ratio $m_p/m_e = 1836$ (-0.01% deviation from CODATA); (ii) the inverse fine-structure constant $\alpha^{-1} = 136.980$ (-0.04% deviation); (iii) the total neutrino mass $\sum m_\nu = 0.09626$ eV (consistent with Planck 2018 upper limit < 0.12 eV); (iv) the neutrino mass spectrum in Normal Hierarchy with baseline $\{0.0206, 0.0206 + 2\delta, 0.0549\}$ eV; (v) the Weinberg angle at the electroweak unification scale $\sin^2\theta_W = 1/3$; (vi) the dark matter and baryonic matter fractions $\Omega_d = 0.2687$, $\Omega_b = 0.0463$ (residual $< 0.3\%$ from Planck); (vii) the Hubble tension ratio $H_0^{\mathrm{late}}/H_0^{\mathrm{early}} = 13/12$; (viii) the CMB scalar spectral index $n_s = 0.96496$ (Planck 2018: 0.9649 ± 0.0042); (ix) a photon dispersion correction coefficient $1/9216$ for $E_\gamma \ll 49$ MeV, with strict $v = c$ for TeV photons. All predictions are parameter-free and will be decisively tested by upcoming experiments (JUNO, DUNE, nEXO, CMB-S4, LHAASO), rendering the framework fully falsifiable within the next decade.

I. INTRODUCTION

Modern physics faces a series of profound theoretical and experimental crises. The Standard Model requires at least 19 free parameters to describe known particle physics phenomena [1], yet cannot explain the origin of dark matter, dark energy, neutrino masses, or why three generations of fermions exist. The standard cosmological model (ΛCDM) faces the severe challenge of the Hubble tension [2–4], and its dark matter particle hypothesis remains unconfirmed after decades of direct detection experiments [5]. The inflationary paradigm, while phenomenologically successful, suffers from the Trans-Planckian problem [6, 7] and relies on finely-tuned scalar field potentials whose fundamental origin remains obscure. The irreconcilable conflict between quantum field theory and general relativity at the Planck scale suggests that the continuous spacetime paradigm may not be the ultimate description of nature.

These dilemmas point to a fundamental question: **Should we abandon the assumption of continuous spacetime and adopt a discrete underlying architecture?**

The Wen Dafang Theory (WDF) presented in this paper is based precisely on this epistemological turn. Its core postulate is remarkably concise: the universe is ultimately constituted of only two elements—**spatial nodes** (the underlying $O_h = 48$ discrete lattice) and **information** (what propagates and organizes between nodes). All physical phenomena—mass, energy, forces, particles—are macroscopic manifestations of different ways in which information organizes on the lattice.

This paper is organized as follows. Section II presents the axiomatic foundation and the core action S_{WDF}. Section III derives the fundamental physical constants. Section IV discusses neutrino dynamics and the mass spectrum. Section V gives the absolute quantization of the Weinberg angle. Section VI establishes the discrete lattice dynamics and derives the photon dispersion prediction. Section VII presents cosmological predictions, including the Hubble tension, dark matter fractions, and the CMB scalar spectral index. Section VIII discusses the relation to existing theories. Section IX outlines the experimental verification program. Section X concludes.

II. WDF AXIOMATIC FOUNDATION AND CORE ACTION

A. Underlying Axioms

WDF is built upon the following irreducible axioms:

Axiom 1 (Discrete Lattice): The underlying structure of physical spacetime is a discrete lattice in three-dimensional space, whose local point group is the full octahedral group O_h, of order 48. This group is the unique discrete symmetry group that simultaneously guarantees seamless 3D tessellation, even parity ($P = +1$) for the gravitational metric tensor, and recovery of Lorentz invariance in the macroscopic limit [9].

Axiom 2 (Information Ontology): There exist no independent "energy," "mass," or "fields" in the universe. All physical entities are macroscopic manifestations of different ways information organizes on the lattice nodes.

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The organization of information is described by k-form dimensional protocols: 0-form (gravity skeleton), 1-form (electromagnetic propagation), 2-form (weak recombination), and 3-form (strong encapsulation).

Axiom 3 (Mass Measure): The electron, as the most fundamental 1-form free propagation state, has its information encapsulation cost defined as the mass measure benchmark $m_e \equiv 1$. The masses of all other particles are integer or fractional multiples of this benchmark.

Axiom 4 (Spacetime Projection): The underlying $O_h = 48$ exists in 4-dimensional spacetime, but its projection onto 3D physical space yields a baseline channel count:

$$N = \frac{O_h}{D_M} = \frac{48}{4} = 12. \tag{1}$$

B. The Ultimate WDF Action

The ultimate action of WDF is given by:

$$S_{\text{WDF}} = \int \frac{d^4x}{\pi^2}\sqrt{-g}\left\{e^{-\alpha\chi^2}(\partial_\mu\phi\partial^\mu\phi)\left[1+(1-\theta)\beta F_{\rho\sigma}F^{\rho\sigma}\right] + \theta\beta F_{\rho\sigma}F^{\rho\sigma}\right\}, \tag{2}$$

where ϕ is the 0-form gravity scalar field, $F_{\rho\sigma}$ is the 1-form electromagnetic field-strength tensor, θ is the Grassmann chiral switch ($\theta^2 = 0$), χ is the local information flux ratio, and $e^{-\alpha\chi^2}$ is the asymptotic decoupling gate. The π^2 factor is the topological compensation Jacobian when translating from the discrete lattice sum $\sum_{\mathbb{K}}$ to the continuous manifold integral $\int d^4x$.

III. RIGID DERIVATION OF FUNDAMENTAL PHYSICAL CONSTANTS

A. Proton-to-Electron Mass Ratio $m_p/m_e = 1836$

Under the on-shell variation $\delta S_{\text{WDF}} = 0$, the system degenerates into a BPS topological saturation state in the macroscopic limit. In this state, the eigen-Hamiltonian on Hilbert space is completely degenerate, proportional to the identity matrix: $H = m_e I$. Mass strictly equals the trace of microstates:

$$M = \text{Tr}(H) = m_e \cdot \dim(\mathcal{H}). \tag{3}$$

The proton, as a 3-form encapsulation state, has its mass composed of two parts:

(i) Volume states: The 3D holographic expansion of the spatial projection channels:

$$M_{\text{bulk}} = N^3 = 12^3 = 1728. \tag{4}$$

(ii) Boundary constraint states: To seal the 3D boundary ∂M and prevent information leakage, a Rank-2 constraint tensor g_{ij} with $d \times d = 3^2 = 9$ independent elements must be applied. The quantum tensor product rule assigns $N = 12$ channels to each element:

$$M_{\text{boundary}} = N \cdot d^2 = 12 \times 9 = 108. \tag{5}$$

The gauge anomaly cancellation condition $\delta_\Lambda S_{\text{bulk}} + \lambda\delta_\Lambda S_{\text{boundary}} = 0$ forces the unique analytical solution $\lambda \equiv 1$. Therefore:

$$\frac{m_p}{m_e} = N^3 + \lambda(N \cdot d^2) = 1728 + 108 = 1836. \tag{6}$$

The CODATA experimental value is 1836.15267343(11) [10], giving a pure geometric residual of -0.01%.

B. Inverse Fine-Structure Constant $\alpha^{-1} \approx 137$

The impedance experienced by electromagnetic information flow on the O_h lattice comprises two contributions:

$$\alpha^{-1} = \frac{D_M}{c_1} + \frac{O_h}{D_M\pi^2} = \frac{4}{c_1} + \frac{12}{\pi^2} = 96\sqrt{2} + \frac{12}{\pi^2} \approx 136.980, \tag{7}$$

where $c_1 = 1/(24\sqrt{2}) \approx 0.02946$ is the topological homology constant. The CODATA experimental value is 137.035999084(21) [10], with a residual of -0.04%.

C. Emergence of $SU(3)$ Gauge Group and 8 Gluons

The scalar kinetic term $(\partial_\mu\phi\partial^\mu\phi)$ defines the symplectic structure of the canonical phase space. Geometric quantization forces the complexification of the real phase space: $\mathbb{R}^3 \to \mathbb{C}^3$. On the $\mathbb{C}^3$ bundle, probability conservation requires unitary holonomy ($U^\dagger U = I$), and volume-preserving topology (color confinement prerequisite) forces $\det(U) = 1$. The unique continuous Lie group satisfying both conditions is $SU(3)$, whose number of generators is $3^2 - 1 = 8$—the 8 gluons emerge automatically from the action without any external hypothesis.

IV. NEUTRINO DYNAMICS AND MASS SPECTRUM

A. Topological Veto of Majorana Mass

Neutrinos originate from the 2-form defect operator $\theta \wedge d$, which fixes left-handed chirality on the 3D lattice. The topological state carries Chern number $C_1 = -1$. Its antiparticle, being the topological inversion, carries $C_1 = +1$. In topological quantum field theory, states with different Chern numbers are absolutely orthogonal in Hilbert space, forcing the Majorana mass coupling integral to vanish identically:

$$\int_M \bar{\nu}_L^C \nu_L \sqrt{-g}\, d^4x \equiv 0. \tag{8}$$

Prediction 1: The event rate of all neutrinoless double beta decay ($0\nu\beta\beta$) experiments will forever be absolute zero.

B. Total Neutrino Mass

The topological leakage of the neutrino from a 2-form transient surface into a 3-form mass state is an instanton tunneling process across a dimensional barrier. The Euclidean path integral yields the suppression factor:

$$\sum m_\nu = m_e \times O_h^{-D_M} = 511{,}000 \times 48^{-4} \approx 0.09626 \text{ eV}. \tag{9}$$

Prediction 2: The total mass of three neutrino generations is 0.09626 eV. The current Planck 2018 upper limit is < 0.12 eV (95% CL) [4]. Future CMB-S4 surveys will test this value decisively.

C. Mass Hierarchy and Spectrum

The neutrino mass matrix is a Rank-1 projector derived from the S_{WDF} variation: $M_\nu \propto c(\mathbf{n} \otimes \mathbf{n}^T)$, with initial spectrum $\{c, 0, 0\}$. First-order perturbation (using the 24 proper rotations of O_h) introduces a leakage rate $1/24$, yielding the bare spectrum $\{c/24, c/24, c\}$. Including the nonlinear compression from 3-form encapsulation ($\eta \approx 0.618$, solved self-consistently from the gap equation), the physical mass spectrum is:

$$\{m_1, m_2, m_3\} = \{0.0206,\ 0.0206 + 2\delta,\ 0.0549\} \text{ eV}, \tag{10}$$

where $\delta \ll m_{1,2}$ is the polarization splitting parameter, generating $\Delta m_{21}^2 > 0$ and the solar neutrino oscillation.

Prediction 3: The neutrino mass ordering must be absolute Normal Hierarchy (NH). The atmospheric mass splitting $\Delta m_{31}^2 \approx 2.5 \times 10^{-3}$ eV^2 is consistent with the current global fit value [11].

D. Maximal CP Violation

In the complexified manifold $\mathbb{C}^3$, the 2-form operator $\theta \wedge d$ corresponds to a purely imaginary antisymmetric matrix. Spectral theorems force its diagonalization to involve eigenroots $e^{\pm i\pi/2}$, endowing the PMNS matrix with a strong preference for maximal CP violation.

Prediction 4: Long-baseline experiments will measure $\delta_{CP} \approx -\pi/2$ ($-90°$). Current data from T2K and NOvA already favor this value [12, 13].

V. ABSOLUTE QUANTIZATION OF THE WEINBERG ANGLE

The weak force is carried by a 2-form geometric plane sliced by $\theta \wedge d$, embedded in the 3-dimensional coordinate representation T_{1u} of the O_h lattice. T_{1u} reduces to a 2-dimensional irreducible representation E_g and a 1-dimensional representation A_{1u}. The W bosons correspond to the E_g excitations (two charged states), while the Z boson corresponds to the full T_{1u} excitation.

On the discrete lattice, the squared mass of a gauge boson is proportional to the number of states in its representation space:

$$\frac{m_W^2}{m_Z^2} = \frac{\dim(E_g)}{\dim(T_{1u})} = \frac{2}{3}. \tag{11}$$

By the standard electroweak relation $m_W = m_Z \cos\theta_W$:

$$\cos^2\theta_W = \frac{2}{3}, \qquad \sin^2\theta_W = \frac{1}{3}. \tag{12}$$

Prediction 5: At the electroweak unification scale, $\sin^2\theta_W = 1/3$. The observed low-energy value $\sin^2\theta_W(m_Z) \approx 0.231$ [1] differs from this bare value due to renormalization group flow driven by the gate factor $e^{-\alpha\chi^2}$ in S_{WDF}.

VI. DISCRETE LATTICE DYNAMICS

A. Hopping Rules and Lattice Constant

On the O_h lattice, the fundamental hopping amplitude of a 1-form information packet has squared modulus $|t_0|^2 = c_1^2 = 1/1152$. Per hop, the internal quantum phase rotates by $\Delta\Phi = 2\pi c_1$ (Hopping Phase Postulate D1).

The electron Compton wavelength is defined as the spatial distance over which the electron's internal phase completes one full 2π cycle:

$$\lambda_e^{\text{WDF}} = \frac{a}{c_1} = 24\sqrt{2} \cdot a. \tag{13}$$

Self-consistency with the Standard Model Compton wavelength $\lambda_e^{\text{SM}} = h/(m_e c)$ in the macroscopic limit yields the absolute lattice constant:

$$a = c_1 \cdot \frac{h}{m_e c} \approx 7.14 \times 10^{-14} \text{ m}. \tag{14}$$

Prediction 6: The characteristic scale of spacetime quantization is on the order of 10^{-13} m, vastly larger than the Planck length (10^{-35} m). This implies that quantum gravity effects may appear far earlier than conventionally expected.

B. Photon Dispersion Correction

A photon is the propagation of a phase disturbance of a 1-form information packet on the lattice. The exact dispersion relation on the discrete lattice is:

$$\omega(k) = \frac{2}{\tau_P} \left| \sin\left(\frac{ka}{2}\right) \right|. \tag{15}$$

In the macroscopic limit $ka \ll 1$ (photon wavelength $\lambda \gg a$), the effective propagation velocity becomes:

$$\begin{aligned} v_\gamma(E_\gamma) &= c\left[1 - \frac{c_1^2}{8}\left(\frac{E_\gamma}{m_e c^2}\right)^2 + \cdots\right] \\ &= c\left[1 - \frac{1}{9216}\left(\frac{E_\gamma}{511 \text{ keV}}\right)^2 + \cdots\right]. \end{aligned} \tag{16}$$

Prediction 7: Photons in the MeV energy range exhibit a vacuum dispersion correction with coefficient 1/9216, testable via GRB time-delay measurements. For TeV photons, the continuum QED description takes over, with strict $v = c$ and zero vacuum dispersion, fully compatible with observations by HESS, MAGIC, and LHAASO.

C. Path Integral and Continuum Limit

The discrete path integral over all hopping trajectories:

$$K(x_f, x_i; T) = \sum_{\gamma: x_i \to x_f} \prod_{\text{hops} \in \gamma} t_{i \to j} \tag{17}$$

converges rigorously to the Feynman path integral in the continuum limit $a \to 0, \tau_P \to 0$ (with $a/\tau_P = c$ fixed), proving that continuous quantum field theory is the exact macroscopic effective description of WDF discrete dynamics.

VII. COSMOLOGICAL PREDICTIONS

A. Hubble Tension

In the early CMB era, the manifold is locked into the baseline 12 information channels. In the late universe, the emergence of dark energy as a global dynamical scalar field (the scale factor $a(t)$) topologically adds one additional macroscopic degree of freedom, forcing an integer channel transition $12 \to 13$:

$$\frac{H_0^{\text{late}}}{H_0^{\text{early}}} = \frac{13}{12} \approx 1.0833. \tag{18}$$

With $H_0^{\text{early}} = 67.4$ km/s/Mpc, we obtain $H_0^{\text{late}} \approx 73.0$ km/s/Mpc, in precise agreement with the Riess et al. measurement [2].

B. Dark Matter and Baryonic Matter Fractions

The baryonic matter fraction corresponds to the successfully rendered information fraction. From the rendering probability in the S_{WDF} path integral:

$$\Omega_b = c_1 \frac{\pi}{2} = \frac{\pi}{48\sqrt{2}} \approx 0.0463. \tag{19}$$

Combined with the Planck measurement $\Omega_\Lambda \approx 0.685$ [4] and the cosmic information budget conservation $\Omega_b + \Omega_d + \Omega_\Lambda = 1$:

$$\Omega_d = 1 - \Omega_\Lambda - \Omega_b \approx 0.2687. \tag{20}$$

This value agrees with the Planck observation 0.268 ± 0.01 to within $< 0.3\%$.

C. CMB Scalar Spectral Index $n_s \approx 0.96$

1. *Resolution of the Trans-Planckian Problem*

The inflationary paradigm suffers from the Trans-Planckian problem: extrapolating backward in time, the physical wavelengths of observed CMB fluctuations would have been far smaller than the Planck length at the onset of inflation, where the continuous spacetime assumption of standard quantum field theory breaks down [6, 7].

In WDF, the underlying O_h discrete lattice possesses a natural UV holographic cutoff at the lattice constant $a \approx 7.14 \times 10^{-14}$ m [Eq. (14)]. No physical scale smaller than a can exist, thereby eliminating the Trans-Planckian catastrophe without invoking any hypothetical inflaton field. The tilt of the scalar spectral index arises directly from the geometric resolution limit of the underlying lattice.

2. *Zero-Order Bare Value: $n_s^{(0)} = 23/24$*

In standard slow-roll inflation, the scalar spectral index is approximately $n_s \approx 1 - 2/N_e$, where N_e is the number of e-folds, typically set to 50–60 to fit observational data [3]. In WDF, the holographic principle and the Bekenstein bound [8] imply that the total entanglement entropy is constrained by boundary topology. The maximum number of independent dynamical degrees of freedom that can participate in the generation of primordial fluctuations is bounded by the group order of the underlying lattice:

$$N_e^{(\text{holographic})} = |O_h| = 48. \tag{21}$$

Substituting this into the spectral index formula yields the bare zero-order prediction:

$$n_s^{(0)} = 1 - \frac{2}{|O_h|} = 1 - \frac{2}{48} = \frac{23}{24} \approx 0.95833. \tag{22}$$

3. Topological Phase Correction

The bare value 23/24 corresponds to the "tree-level" discrete lattice result. In the continuum limit, the holographic projection of the O_h lattice onto the 2-sphere S^2 of the CMB sky introduces a topological phase correction. The Gaussian gating operator $e^{-\alpha\chi^2}$ in S_{WDF} [Eq. (2)], when integrated over the closed boundary manifold S^2, yields a topological invariant by the Gauss-Bonnet theorem [14]. Since the Euler characteristic of S^2 is $\chi_E = 2$, the boundary integral contributes an absolute topological charge independent of the local parameter α:

$$\Delta\gamma = \frac{1}{|O_h|}\oint_{\partial M} \mathcal{G}(\chi)\, d\Omega = \frac{1}{|O_h|}\frac{\chi_E}{2\pi} = \frac{1}{\pi|O_h|} = \frac{1}{48\pi}. \tag{23}$$

This positive contribution counteracts part of the discrete dissipation, yielding the complete analytical prediction:

$$n_s = 1 - \frac{2}{48} + \frac{1}{48\pi} = 1 - \frac{1}{24} + \frac{1}{48\pi} \approx 0.96496. \tag{24}$$

Prediction 8: The CMB scalar spectral index is $n_s = 0.96496$. The Planck 2018 measurement yields $n_s = 0.9649 \pm 0.0042$ [3], in remarkable agreement with the WDF value. The deviation from unity arises entirely from the finite group order of the discrete lattice and the topological phase of the holographic boundary, requiring no inflaton potential, slow-roll parameters, or dark matter input.

VIII. RELATION TO EXISTING THEORIES

It is important to clarify how WDF relates to established physical frameworks and to other alternative approaches.

A. Relation to the Standard Model and General Relativity

WDF does not reject the Standard Model or General Relativity as phenomenologically valid descriptions. Rather, it provides an underlying microscopic foundation from which these effective theories emerge. In the continuum limit $a \to 0$, the discrete path integral of WDF rigorously converges to the Feynman path integral of quantum field theory [Eq. (17)], and the S_{WDF} action reduces to the Einstein-Hilbert action plus gauge field kinetic terms in the low-curvature limit. The Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ partially emerges from the geometric quantization procedure, with the $SU(3)$ sector derived in Section III.C. The electroweak $SU(2) \times U(1)$ structure and the origin of the Higgs mechanism will be addressed in a forthcoming work.

B. Comparison with Modified Newtonian Dynamics (MOND)

Like MOND [15], WDF explains galactic rotation curves without invoking particle dark matter. However, the underlying mechanism is fundamentally different: whereas MOND postulates a modification of Newtonian dynamics below a critical acceleration a_0, WDF attributes the apparent excess gravity to the information rendering bottleneck on the O_h lattice—the "unrendered information skeleton" acts as an invisible gravitational source. The critical acceleration a_0 in WDF is not a free parameter but is rigidly derived from geometric constants: $a_0 = c_1 \cdot H_0 \cdot c$ (dimensional analysis confirmed in Section III of the WDF field theory exposition).

C. Comparison with Loop Quantum Gravity and String Theory

Like loop quantum gravity (LQG) [16] and string theory [17], WDF posits a discrete microstructure of spacetime. However, WDF differs sharply in its predictions. While LQG and string theory generally predict linear modifications to photon dispersion relations ($\Delta v/c \sim E/E_{\text{Planck}}$), WDF predicts a quadratic correction with a distinct coefficient 1/9216 and a characteristic scale ($\sim 10^{-13}$ m) vastly larger than the Planck length. This makes WDF uniquely testable against other quantum gravity candidates using GRB time-delay measurements.

IX. EXPERIMENTAL VERIFICATION PROGRAM

The rigid predictions of WDF will be decisively tested within the next decade:

X. CONCLUSION

We have presented a predictive phenomenological framework—Wen Dafang Theory—based on discrete

TABLE I. WDF Rigid Predictions and Experimental Tests

Prediction	Value	Experiment	Timeline
Neutrino mass ordering	Absolute NH	JUNO	2027–2028
δ_{CP}	$\approx -90°$	DUNE, Hyper-K	2028–2030
$\sum m_\nu$	0.09626 eV	CMB-S4, Simons Obs.	2027–2030
$0\nu\beta\beta$ event rate	$\equiv 0$	nEXO, LEGEND	2028–2032
$\sin^2\theta_W$ (high scale)	1/3	FCC, CEPC	2035+
Photon dispersion	1/9216	LHAASO, CTA	2025–2030
Scalar spectral index n_s	0.96496	Planck (confirmed)	Measured

$O_h = 48$ lattice geometry and k-form information dimensional protocols. The theory provides zero-parameter rigid derivations of: the proton-to-electron mass ratio (1836), the inverse fine-structure constant (≈ 137), the total neutrino mass (0.09626 eV), the neutrino mass hierarchy and spectrum, maximal CP violation ($\delta_{CP} \approx -90°$), the Weinberg angle at the electroweak scale ($\sin^2\theta_W = 1/3$), the Hubble tension ratio (13/12), the dark matter and baryonic matter fractions, the CMB scalar spectral index (0.96496), and a photon dispersion correction (1/9216).

All nine predictions are parameter-free, mutually consistent, and will be decisively tested by experiments within the next decade. The vitality of WDF lies in its extreme falsifiability: any significant deviation from its predictions would constitute a fatal refutation. We invite the global physics community to independently verify these predictions and to subject the theory to rigorous experimental scrutiny.

Data Availability Statement:
No experimental data were generated or analyzed in this theoretical work. All numerical predictions are derived from first principles without recourse to external data fitting. The theoretical values presented can be verified by direct computation from the formulas provided in the text.

Conflict of Interest Statement:
The author declares no competing financial or personal interests that could have influenced this work. This research is conducted as an independent investigation without any funding source or institutional affiliation.

Acknowledgments:
This is an independent research work. The author received no specific funding for this study.

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